Week 10 Tutorial – Time Series – MEMO

Question 1

a) Random variation.

b) k = 4 would be appropriate. Our data is recorded in quarters and there are 4 quarters in one year.

c) A = 70.500
B = 68.000
C = 71.000
D = 69.265

Question 2

a) There is a Random component present in the data series.

b) Either: There is no loss of data **OR**: Exponential smoothing models take all the data into account.

c)

Month	t	Sales (Y_t)	Exp$(0.4)_t$
April	1	18	18.00
May	2	22	19.60
June	3	19	19.36
July	4	18	18.82
August	5	20	19.29
September	6	16	17.97
October	7	21	19.18
November	8	24	21.11
December	9	16	19.07

Question 3

a) There is random variation present in the time series.

b) A = 47.75
B = 45.88
C = 43.75
D = 51.00
E = 50.09
F = 45.84

c)

Year	t	Sales	Exp(0.5)	MA(3)
2003	1	51	51	
2004	2	38	44.5	47.33
2005	3	53	48.75	46
2006	4	47	47.88	51.67
2007	5	55	51.44	48.67
2008	6	44	47.72	45.33
2009	7	37	42.36	43
2010	8	48	45.18	42.33
2011	9	42	43.59	47.33
2012	10	52	47.8	

d) A smaller value of k results in a less-smoothed series.

e) A larger value for α results in a less-smoothed series.

f) Advantages: It is easy and quick to apply.
Disadvantages: You lose data values; once an observation falls outside the window being used to calculate a moving average, it is never considered again.

g) It does not result in the loss of any data values; and each smoothed value takes all the observed values in previous time periods into account.

h) Moving average smoothing typically is not used to forecast future values of a time series. However, since there is only random variation present here, it could be used. The forecasts from the simple exponential smoothing model would be seen to more reliable here though.

Question 4

a) Yes. There is no evidence in the time series to suggest that there is a trend **or** a seasonal component.

b) It does not result in a loss of any data values.
Each smoothing calculation is based on all previously observed values.

c)

Year	Demand	Exp(0.2)
2003	42	42.000
2004	43	42.200
2005	42	42.160
2006	45	42.728
2007	44	42.982
2008	45	43.386
2009	44	43.509
2010	42	43.207
2011	42	42.966
2012	43	42.972

d) A larger **α** would have resulted in less smoothing of the original data.

e) 42.972 ≈ 43 000 toothpaste tubes.

Question 5

M

a) There is random variation in this time series.
b) This time series is plausibly stationary as there is no long-term trend and the variance looks relatively constant.
c) No transformations are necessary as the time series appears stationary.
d) The naïve or average method would be appropriate here as there is no trend or seasonality.

N

a) There is an upward linear trend and random variation present in the time series.
b) This time series is not stationary as there is a long-term upward linear trend, which means that the mean is changing with time. The variance is not changing with time.
c) First-order differencing at lag p = 1 would be an appropriate transformation to apply to this time series to result in a stationary series
d) The drift method would be the most appropriate due to the presence of the linear trend

O

a) There is an upward linear trend and random variation present in the time series.
b) This time series is not stationary as there is a long-term upward linear trend (so the mean is changing with time. The variance is not changing with time.
c) First-order differencing at lag p = 1 would be an appropriate transformation to apply to this time series to result in a stationary series.
d) The drift method would be the most appropriate due to the presence of the linear trend.

P

a) There is an upward non-linear trend, seasonality and random variation present in the time series.
b) This time series is not stationary as there is a long-term upward non-linear trend and seasonality (so the mean is changing with time) and increasing variance.
c) A variance stabilizing transformation (e.g. taking the logarithm of the series) and then first-order differencing at lag p = 12 (of the logged data) would be appropriate transformations to apply to this time series to result in a stationary series.
d) A seasonal naïve model would be the best short-term simple forecasting model (say for a year's worth of forecasts). The drift method would be the best medium-term simple forecasting model as it would account for the trend. However, neither of these would be appropriate for forecasts far into the future as the trend is not linear.

Q

a) There is random variation in this time series. No trend or seasonality.
b) This time series is plausibly stationary as there is no long-term trend, and the variance looks relatively constant.
c) No transformations are necessary as the time series appears stationary.
d) Average or naïve method as there is no trend or seasonality.

R

a) There is random variation in this time series. There is also an oscillating pattern in the time series (not seasonality, but still a pattern).
b) The variance does not appear to be constant. Whilst there is no long-term trend in the series, we cannot conclude that the series is stationary.
c) Some form of variance stabilizing transformation.
d) Average method – there is no trend or seasonality, but the values are oscillating about the mean value of the series. This would result in the naïve method giving very inaccurate forecasts.

S

a) There is random variation in this time series. No trend or seasonality.
b) The variance and mean appear relatively constant. This time series is stationary.
c) No transformations would be applied.
d) Average or naïve method as there is no trend or seasonality present in the time series

T

a) There is random variation and seasonality in this time series. Note that this is actually an ACF plot. We will see that when there is a seasonal pattern in the ACF, it means that there is seasonality in the time series.
b) Whilst there is no long-term trend (we will learn what an ACF for a time series with seasonality and trend looks like), the seasonal variation means that the mean is not constant through time.
c) Probably a variance-stabilizing transformation and first order differencing at lag $p = 12$.
d) Seasonal naïve method due to strong seasonality and no trend present in the time series.

U

a) There is an upward linear trend (note – the trend is potentially non-linear) and random variation present in the time series.
b) This time series is not stationary as there is a long-term upward linear trend, which means that the mean is changing with time. The variance is not changing with time.
c) First order differencing at lag $p = 1$ would be an appropriate transformation to apply to this time series to result in a stationary series.
d) Drift method due to the upwards linear trend.

Question 6

a) Forecasted value:

$$\widehat{y_{t+1}} = y_t + (1)(Drift) = 813.67 + (1)(0.4213) = 814.0913$$

80% prediction interval:

$$814.0913 \pm 1.28(8.7285) = 814.0913 \pm 11.1725 = [802.9188; 825.2638]$$

b) Forecasted value:

$$\widehat{y_{t+5}} = y_t + (5)(Drift) = 813.67 + (5)(0.4213) = 815.7765$$

90% prediction interval:

$$815.7765 \pm 1.64(8.7285) = 815.7765 \pm 14.3147 = [801.4618; 830.0912]$$

c) Forecasted value:

$$\widehat{y_{t+12}} = y_t + (12)(Drift) = 813.67 + (12)(0.4213) = 818.7256$$

60% prediction interval:

$$818.7256 \pm 0.84(8.7285) = 818.7256 \pm 7.3319 = [811.3936; 826.0575]$$

d) These metrics measure the retrospective forecast accuracy of a fitted model. The smaller the value (closer to 0) of the metric, the better the retrospective forecast accuracy. You choose the model that has the lower MAD or MSE as your forecasting model.

e)

	Mod1	Mod2	Mod3
MAD	10.99	9.73	9.76

f) Mod2, because it has the lowest MAD value.

Question 7

a) Seasonality, potentially a slight upwards trend and random variation.
b) Yes, at lag 12 – every 12 months the same pattern repeats itself (seasonality).
c) No, the mean of this time series is not constant due to the presence of seasonality (and potentially a trend).
d) Yes, first order differencing at lag 12 to remove the seasonality and the trend. There doesn't appear to be a problem with changing variance.
e)

The Ljung-Box test is the appropriate test to use here.

H0: There is no autocorrelation up to lag 3 in the time series
H1: There is autocorrelation in at least one lag up to lag 3 in the time series

Test statistic:

(note that there are 56 total months in the dataset – look at the time series plot)

$$Q_{LB}(m) = T(T+2) \times \sum_{h=1}^{m} \frac{\hat{\rho}(h)^2}{T-h}$$

$$Q_{LB}(3) = 56(58) \times \sum_{h=1}^{3} \frac{\hat{\rho}(h)^2}{56-h}$$

$$= 3248 \times \left(\frac{(0.091)^2}{55} + \frac{(-0.225)^2}{54} + \frac{(-0.116)^2}{53} \right)$$

$$= 4.359$$

From the Chi-squared distribution table, using 3 degrees of Freedom, the p-value of this test statistic > 0.1. There is thus insufficient evidence to reject the null hypothesis, and we conclude that there is no autocorrelation up to lag 3 in the time series.